

Quantum Molecular Dynamics

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Executive summary

This work studies a quantum many-body simulation method based on the work from Professor Fantoni R. *Quantum Molecular Dynamics*. The density matrix evolves in imaginary-time according to the Bloch equation, starting from a delta-function fiducial state. Klauder's continuous representation is then used to parametrize the quantum evolution through classical labels for positions and momenta. By restricting the quantum action to this family of states, one obtains a classical action and Hamilton equations for the labels. Quantum corrections arise, but they reduced by choosing a sharply localized state, while the divergent kinetic contribution is formally removed through a phase term. The construction is extended to identical particles using symmetrization or antisymmetrization. Finally, the diagonal thermal density matrix is used to compute thermal averages through Monte Carlo integration.

Notation:

Results are in red.

Assumptions are in blue.

Proofs or calculations are in gray.

Q, P are operators, even if it is not always specified the $\hat{}$.

1 Introduction

1.1 System under examination

Let a system be made of N particles, each with mass m . Their state is described by position and momentum, neglecting spin, Q and P , with

$$Q = (q_1, \dots, q_N), \quad \hat{P} = (\hat{p}_1, \dots, \hat{p}_N) = -i\hbar(\nabla_{q_1}, \dots, \nabla_{q_N}).$$

According to quantum mechanics, the behaviour of this system is described by the Schrödinger equation,

$$i\hbar \frac{\partial}{\partial t_r} \psi(Q, t_r) = \hat{H} \psi(Q, t_r),$$

for which one has to find the Hamiltonian

$$\hat{H}(\hat{Q}, \hat{P}) = \hat{K} + \hat{V} = \frac{\hat{P}^2}{2m} + V(\hat{Q}).$$

1.2 The role of the potential

The interaction is assumed to be pairwise $V(Q) = \sum_{i < j} v(|q_i - q_j|)$. This assumption implies that V is a ‘c-number’ in the position representation. A ‘c-number’ is defined as a ‘classical’ number, meaning that, because it depends only on Q , it acts by multiplication:

$$[V(\hat{Q})\psi](Q) = V(Q)\psi(Q).$$

V may be not quadratic. This implies that Klauder’s continuous representation for such a Hamiltonian falls in the *inexact category*. To define Klauder’s representation uses a restricted family of states

$$\mathcal{S} = \{\phi[Q, P, \alpha; \cdot]\}.$$

The representation is exact if the quantum evolution ‘stays inside’ $\mathcal{S}$, i.e. if

$$e^{-\tau \hat{H}} \phi_0 = \phi[Q(\tau), P(\tau), \alpha(\tau); \cdot].$$

For quadratic Hamiltonians, this closure does hold. For a generic non-quadratic potential, the evolution distorts the state in a way that cannot be represented only by changing Q , P , and α . Thus the representation is inexact.

2 Quantum mechanics and Dirac-Frenkel principle

Before diving into the results of the paper and the design of the algorithm, I try to give a brief introduction about some fundamental concepts.

Dirac-Frenkel principle is a variational principle used in approximating solutions of PDEs in a manifold, when computing the solution on the whole Hilbert space has a too high cost.

Principle 1 Dirac–Frenkel variational principle. Let $\mathcal{M} \subset \mathcal{H}$ be a manifold of trial states, and let $\psi(t) \in \mathcal{H} \forall t$ satisfy the evolution problem

$$\dot{\psi}(t) = F(\psi(t)) \quad \wedge \quad \psi(0) = \psi_0$$

Then, the approximate evolution $u(t) \approx \psi(t)$, $u(t) \in \mathcal{M}$ is determined by constructing $\dot{u}(t) \in T_{u(t)}\mathcal{M}$ such that the Galerkin condition holds:

$$\forall v \in T_{u(t)}\mathcal{M} : \quad \langle v, \dot{u}(t) - F(u(t)) \rangle = 0.$$

with $u(0) = \psi_0$. $T_{u(t)}\mathcal{M}$ is the space which is tangent to the manifold in $u(t)$.

This can be applied to the Schroedinger equation as well, with $F = H/i\hbar$. However, if the **choice** is $\mathcal{M} = \mathcal{H}$, the approximating manifold coincides with all the Hilbert space, then one obtains the Schroedinger equation. A variational formulation follows naturally for quantum dynamics:

Principle 2 (Quantum action principle) The evolution of the wave function makes the quantum action functional stationary:

$$\delta I[\psi] = 0.$$

With

$$\begin{aligned} I[\psi] &= \int dt \left\langle \psi(t) \left| i\hbar \frac{d}{dt} - \hat{H} \right| \psi(t) \right\rangle \\ &= \int dt \left[i\hbar \langle \psi(t) | \dot{\psi}(t) \rangle - \langle \psi(t) | \hat{H} | \psi(t) \rangle \right] \end{aligned}$$

Why is this important? It is used in the paper and in Klauder's work to show the equivalence with classical dynamics. In classical mechanics the minimum action principle is formulated as follows.

Principle 3 Hamilton's variational principle. The classical trajectory $(q(t), p(t))$ makes the action stationary:

$$\delta S[q, p] = 0,$$

$$S[q, p] = \int_{t_1}^{t_2} dt \left[p(t) \dot{q}(t) - H(q(t), p(t), t) \right].$$

Then, with $\delta q(t_1) = \delta q(t_2) = 0$, this yields the Euler-Lagrange equations

$$\dot{q} = \frac{\partial H}{\partial p},$$

$$\dot{p} = -\frac{\partial H}{\partial q}.$$

3 Reduction of quantum action functional to classical action functional

Let us show the reduction of quantum action functional to classical action functional, from the study of a one-dimensional particle. Assumptions: non-relativistic, one-dimension, no interactions, use quantum mechanics.

The particle's state is then described by position $\hat{Q}$ and momentum $\hat{P}$.

Define the *overcomplete family of states* (OFS) X , as a parametrized manifold w.r.t. the continuous parameters $p, q \in R$

$$X := \left\{ \psi \in \mathcal{H} \mid \exists \psi_0 \in \mathcal{H} : \psi = \psi(q, p) = e^{-iq\hat{P}/\hbar} e^{ip\hat{Q}/\hbar} \psi_0 \right\}.$$

The proof of X being an OFS is omitted but it comes from three postulates.

The following conditions are imposed so that the labels p, q represent the expected values of momentum and position for each element in X :

$$(\psi(p, q), \hat{P}\psi(p, q)) = p \quad \wedge \quad (\psi(p, q), \hat{Q}\psi(p, q)) = q.$$

which is equivalent by construction to impose

$$\psi_0^* \hat{P} \psi_0 = 0 \quad \wedge \quad \psi_0^* \hat{Q} \psi_0 = 0. \quad (0)$$

Notice that the factor $e^{ip\hat{Q}/\hbar}$ shifts the wavefunction in momentum space by p . Then $e^{-iq\hat{P}/\hbar}$ shifts it in position space by q . Thus, $|p, q\rangle$ is the fiducial state $|0\rangle$ displaced in phase space.

Introduce time dependent evolution of the wave function by stationarity of the action functional

$$I = \int dt \left[i\hbar \left\langle \psi(t) \left| \frac{d\psi(t)}{dt} \right\rangle - \langle \psi(t) | \hat{H} | \psi(t) \rangle \right].$$

and by letting q, p depend on time $\rightarrow q(t), p(t)$, so that they parametrize the evolution restricted to X .

Indeed, restrict the dynamics to the unit vectors in X through a Dirac-Frenkel principle

$$\left\langle v \left| i\hbar \frac{d}{dt} \psi(t) - \hat{H} \psi(t) \right\rangle = 0 \quad \forall v \in \mathbf{T}_{\psi(t)} X$$

Then it can be proved as follows that the restricted dynamics is equivalent to the evolution given by classical action functional with hamiltonian $H(p, q) = H_{\text{classical}}(p, q) + \epsilon(\hbar; \psi_0; p, q)$,

$$I = \int dt [p\dot{q} - H(p, q)].$$

Proof:

1st step: The conditions imposed before on the interpretation of p and q lead to the kinematical form

$$i\hbar \langle p, q | d | p, q \rangle = p dq + d\gamma,$$

where $d\gamma$ is an exact differential; therefore, it produces only a boundary term in the action and can be absorbed in a phase convention. Choosing this convention gives

$$i\hbar\langle p, q|d|p, q\rangle = p dq.$$

2nd step: Combine the last result with $q(t), p(t)$:

$$i\hbar\left\langle\psi(t)\left|\frac{d}{dt}\right|\psi(t)\right\rangle = p(t)\dot{q}(t).$$

3rd step: Define the classical Hamiltonian as the expectation value of the quantum Hamiltonian:

$$H(p, q) = \langle p, q|\hat{H}(\hat{P}, \hat{Q})|p, q\rangle.$$

4th step: Define D as the displacement operator, the one constructing X

$$D\psi(Q, P) = e^{-iq\hat{P}/\hbar}e^{ip\hat{Q}/\hbar}\psi(Q, P) = \psi(Q + q, P + p)$$

Then, one obtains

$$\begin{aligned}\hat{D}^\dagger(p, q)\hat{P}\hat{D}(p, q) &= \hat{P} + p, \\ \hat{D}^\dagger(p, q)\hat{Q}\hat{D}(p, q) &= \hat{Q} + q,\end{aligned}$$

one obtains

$$H(p, q) = \langle\psi(q, p)|\hat{H}(\hat{P}, \hat{Q})|\psi(q, p)\rangle = \langle\psi_0|\hat{D}^\dagger(p, q)\hat{H}(\hat{P}, \hat{Q})\hat{D}(p, q)|\psi_0\rangle = \langle\psi_0|\hat{H}(\hat{P}+p, \hat{Q}+q)|\psi_0\rangle.$$

5th step: Separate the classical contribution from the quantum correction ϵ :

$$\langle\psi_0|\hat{H}(\hat{P} + p, \hat{Q} + q)|\psi_0\rangle = H_{\text{cl}}(p, q) + \epsilon(\hbar; \psi_0; p, q).$$

with $H_{\text{cl}}(p, q) = \frac{p^2}{2m} + V(q)$. Therefore,

$$H(p, q) = H_{\text{cl}}(p, q) + \epsilon(\hbar; \psi_0; p, q).$$

6th step: Substitute the kinematical and dynamical terms into the action:

$$I[p, q] = \int dt \left[i\hbar\left\langle\psi(t)\left|\frac{d}{dt}\right|\psi(t)\right\rangle - \langle\psi(t)|\hat{H}|\psi(t)\rangle \right].$$

Using

$$i\hbar\left\langle\psi(t)\left|\frac{d}{dt}\right|\psi(t)\right\rangle = p(t)\dot{q}(t),$$

and

$$\langle\psi(t)|\hat{H}|\psi(t)\rangle = H(p(t), q(t)),$$

we obtain

$$I[p, q] = \int dt [p(t)\dot{q}(t) - H(p(t), q(t))].$$

7th step: Substituting the decomposition of the Hamiltonian gives

$$I[p, q] = \int dt [p(t)\dot{q}(t) - H_{cl}(p(t), q(t)) - \epsilon(\hbar; \psi_0; p(t), q(t))].$$

This means the dynamics can be approximated by the one on c-numbers given by a classical hamiltonian

$$I_{quantum}[\psi] \approx I[p, q] = \int dt [p(t)\dot{q}(t) - H_{cl}(p(t), q(t)) - \epsilon(\hbar; \psi_0; p(t), q(t))] \quad (1)$$

$$\approx \int dt [p(t)\dot{q}(t) - H_{cl}(p(t), q(t))]. \quad (2)$$

Interpretation: the dynamics of the system described by quantum mechanics, under the postulate of Schroedinger equation, is approximatively described by a classical dynamics($H_{classical}$) plus an additional (error) term ϵ . The classical coordinates q, p represent the expected values of the operators $\hat{Q}, \hat{P}$. Neglecting the error term, one obtains the Euler-Lagrange equations for q, p . The result holds when the state is restricted to the chosen OFS, the fiducial vector is normalized and centred :

$$\langle \psi_0, \psi_0 \rangle = 1, \quad \langle \psi_0, \hat{Q}\psi_0 \rangle = 0, \quad \langle \psi_0, \hat{P}\psi_0 \rangle = 0.$$

and the variational evolution remains a good approximation to the full quantum evolution. The evolution is exact only when the exact evolution remains inside the chosen family of states, as it can for quadratic Hamiltonians.

The issue is this: given a quantum theory on Hilbert space, what does it mean to extract a classical system from it, with physically meaningful coordinates and a Hamiltonian? Klauder's answer is: choose a phase-space surface of quantum states $|p, q\rangle$, interpret p, q operationally as mean values, and restrict the quantum action to that surface. Thus, it gives a closed, canonical classical-looking dynamics with quantum-fluctuation corrections included inside $H(p, q)$.

4 Algorithm design

Let's start with a simplified case, then extended it to a more general one for fermions and bosons.

Let a system be composed of N distinguishable particles. Each possible state is described by the density matrix (in position representation)

$$\rho(Q, Q', t_r)$$

with $Q = (q_1, \dots, q_N)$, $\hat{P} = -i\hbar\nabla_Q$.

First, let us consider only the diagonal term, that we will define as $\psi(Q, t_r) := \langle Q | \rho(Q, Q, t_r) | Q \rangle$. The reduction is performed in imaginary time by Wick rotation $t_r = -i\hbar t$, and $\psi(Q, t_r) \rightarrow \phi(Q, t)$ where t_r is real time and t is imaginary time. Therefore, the real-time Schrödinger equation becomes the imaginary-time Bloch equation

$$i\hbar \frac{\partial \psi}{\partial t_r} = \hat{H}\psi \quad \rightarrow \quad \frac{\partial \phi}{\partial t} = -\hat{H}\phi.$$

Thus, evolution is described by the minimization of the quantum action functional in imaginary-time t .

$$I[\phi] = \int dt \left[\left(\phi, \frac{\partial \phi}{\partial t} \right) + (\phi, \hat{H} \phi) \right].$$

with $\hat{H}(\hat{Q}, \hat{P}) = \hat{K}(\hat{P}) + \hat{V}(\hat{Q}) = \frac{\hat{P}^2}{2m} + V(\hat{Q})$.

Assume: (i) $V(\hat{Q}) = \sum_{i < j} v(|q_i - q_j|)$, pairwise position-dependent interactions; the particles are (ii) **distinguishable**; interpretation (0) that was (iii) $(\phi_0, \hat{Q} \phi_0) = 0 \quad \wedge \quad (\phi_0, \hat{P} \phi_0) = 0$; normalized fiducial vector (iv) $(\phi_0, \phi_0) = 1$.

Introduce the c-number labels q_i, p_i for each particle, such that the state is represented as

$$\psi(Q, t) = e^{-t\hat{H}} \psi_0(Q)$$

while in the reduced dynamics

$$\phi[q, p; Q] = e^{-i\alpha/\hbar} e^{-iq \cdot \hat{P}/\hbar} e^{ip \cdot \hat{Q}/\hbar} \phi_0(Q).$$

From the **result** in section 3, the quantum action reduces to

$$I[q, p] = \int dt \left[\sum_{i=1}^N p_i \cdot \dot{q}_i + \dot{\alpha} - H(q, p) \right].$$

The effective Hamiltonian is

$$H(q, p) = -i\hbar \left(\phi_0, \hat{H}(q + \hat{Q}, p + \hat{P}) \phi_0 \right).$$

Replacing the operator arguments by the classical labels gives

$$H(q, p) = -i\hbar \left[\hat{H}(q, p) + \epsilon(\hbar; \phi_0; q, p) \right]$$

where the correction depends on the fiducial vector and on the classical labels.

4.1 How to handle the dynamics

The goal of this section is to make the correction term ϵ as small as possible.

(A) Quantum correction term ϵ

It can be shown that

$$\epsilon = \frac{(\phi_0, \hat{P}^2 \phi_0)}{2m} + \left(\phi_0, [V(q + \hat{Q}) - V(q)] \phi_0 \right) =: c + w(q).$$

Showing that $\epsilon = c + w(q)$: For $\hat{H} = \hat{K}(\hat{P}) + \hat{V}(\hat{Q})$, we have $\hat{H}(q + \hat{Q}, p + \hat{P}) = \hat{K}(p + \hat{P}) + V(q + \hat{Q})$.

Separating the classical terms from the fluctuation terms gives a corrective term

$$H(q, p) = -i\hbar \left[\hat{K}(p) + V(q) + c + w(q) \right],$$

where

$$c = (\phi_0, [\hat{K}(p + \hat{P}) - \hat{K}(p)] \phi_0),$$

and

$$w(q) = (\phi_0, [V(q + \hat{Q}) - V(q)] \phi_0).$$

But c can be rewritten noticing that $\hat{K}(p + \hat{P}) = \frac{p^2}{2m} + \frac{p \cdot \hat{P}}{m} + \frac{\hat{P}^2}{2m}$ and using the fiducial-vector condition $(\phi_0, \hat{P} \phi_0) = 0$ which implies $(\phi_0, \frac{p \cdot \hat{P}}{m} \phi_0) = \frac{p}{m} \cdot (\phi_0, \hat{P} \phi_0) = 0$. Thus $c = \frac{(\phi_0, \hat{P}^2 \phi_0)}{2m}$ and $H(q, p) = -i\hbar \left[\frac{p^2}{2m} + V(q) + c + w(q) \right]$.

(B) Potential energy correction $w(q)$ The term $(\phi_0, [V(q + \hat{Q}) - V(q)] \phi_0)$ is approximated to zero thanks to the following ansatz about the fiducial vector

$$\phi_0(Q) \simeq \delta_{Dirac}(Q) \Rightarrow w(q) \simeq 0$$

However, this causes the kinetic term to diverge as shown in the following point (C).

(C) Kinetic term c The factor $e^{-i\alpha/\hbar}$ is a phase factor. In real-time quantum mechanics it is a global phase and does not change the probability distribution. It is used in the present imaginary-time construction, to remove the kinetic term that could otherwise blow to infinity, that is $c = \frac{(\phi_0, \hat{P}^2 \phi_0)}{2m}$.

Then the phase factor permits to **formally** delete such divergence which could cause problems numerically, by the following choice:

$$\alpha = -i\hbar c t, \quad \dot{\alpha} = -i\hbar c.$$

1. Let us show the divergence of c for ϕ_0 sharply localized:

Let $c = \frac{(\phi_0, \hat{P}^2 \phi_0)}{2m}$. Assume $(\phi_0, \hat{P} \phi_0) = 0$. Then $(\phi_0, \hat{P}^2 \phi_0) = (\Delta P)^2$. Therefore, $c = \frac{(\Delta P)^2}{2m}$. For the position operator, $(\phi_0, \hat{Q}^2 \phi_0) \rightarrow 0$ and, if $(\phi_0, \hat{Q} \phi_0) = 0$, then $(\Delta Q)^2 \rightarrow 0$. The Heisenberg uncertainty relation reads $\Delta Q, \Delta P \geq \frac{\hbar}{2}$. Hence, $\Delta P \geq \frac{\hbar}{2\Delta Q}$. Consequently, $c = \frac{(\Delta P)^2}{2m} \geq \frac{\hbar^2}{8m(\Delta Q)^2} \rightarrow \infty$ as $\Delta Q \rightarrow 0$. Thus the precise statement is $(\phi_0, \hat{Q}^2 \phi_0) \rightarrow 0 \implies (\phi_0, \hat{P}^2 \phi_0) \rightarrow \infty$ under the stated assumptions.

2. Simplification thanks to phase factor:

$$\begin{aligned} I[q, p] &= \int dt \left[\sum_{i=1}^N p_i \cdot \dot{q}_i + \dot{\alpha} + i\hbar \left(\frac{p^2}{2m} + V(q) + w(q) \right) + i\hbar c \right] \\ &= \int dt \left[\sum_{i=1}^N p_i \cdot \dot{q}_i - i\hbar c + i\hbar \left(\frac{p^2}{2m} + V(q) + w(q) \right) + i\hbar c \right] \\ &= \int dt \left[\sum_{i=1}^N p_i \cdot \dot{q}_i + i\hbar \left(\frac{p^2}{2m} + V(q) + w(q) \right) \right]. \end{aligned}$$

4.2 Result

From the previous procedure, we define $H_{cl} := \left(\frac{p^2}{2m} + V(q) + w(q) \right)$ and we obtain

$$I[q, p] \simeq \int dt \left[\sum_{i=1}^N p_i \cdot \dot{q}_i - H_{cl}(q, p) \right].$$

Hence, the (approximate) equations of motion are Hamilton's equations:

$$\dot{q}_i = \frac{\partial H_{cl}(q, p)}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H_{cl}(q, p)}{\partial q_i}.$$

4.3 Generalization to off diagonal terms

So far, the description has implicitly assumed that we were describing a density matrix with respect to only one variable Q . That is also relevant in the assumption (0), which actually coincides with looking at only the first column of the density matrix. This permits to write the formalism in an easier way, now we generalize.

Let $\phi_0(Q, Q') = \delta(Q - Q')$, this assumption incorporates the previous one $\phi_0(Q, 0) = \delta(Q - 0)$, while assumption (0) becomes

$$\langle \phi_0 | \hat{P} | \phi_0 \rangle = P' \quad \wedge \quad \langle \phi_0 | \hat{Q} | \phi_0 \rangle = Q'.$$

Under this situation, the classical action functional becomes

$$I'[Q, P] = \int \left(\sum_{i=1}^N [(P_i + p'_i) \cdot \dot{Q}_i - \dot{P}_i \cdot q'_i] + \dot{\alpha} - H(Q, P) \right) dt = I[Q, P] + \mathbf{f} + \mathbf{g}$$

where $\mathbf{f}$ and $\mathbf{g}$ are two constant terms, thus they do not play any role in the stationarity of the functional, then no role in the dynamics. The change appears in the hamiltonian as

$$H_{cl} = \dot{\alpha} - H(Q, P) = i\hbar [\hat{K}(P) + \hat{V}(Q' + Q)].$$

4.4 From the c-numbers to observables

The goal is to compute observables as averages over the thermal density matrix, that is at [thermal equilibrium](#) with an external bath at temperature T .

In imaginary time evolution

$$\phi(Q, t) = e^{-t\hat{H}} \phi_0(Q)$$

with the density operator denoted as $\rho(t) = e^{-t\hat{H}}$.

To obtain the density matrix at equilibrium, we evolve the system from $t=0$ to $t=\beta = \frac{1}{k_B T}$:
 $\phi(Q, Q', \beta) = \rho(\beta) \phi_0(Q, Q')$.

Take an observable which has a [diagonal coordinate representation](#), so that we can compute

$$\langle O \rangle = \frac{\int_{\Omega^N} O(Q, P) \phi(Q, Q, \beta) dQ}{\int_{\Omega^N} \phi(Q, Q, \beta) dQ}.$$

5 Identical particles and the fermion sign problem

5.1 Fermions sign problem

So far, the particles have been assumed to be distinguishable. This assumption is not valid for identical fermions, neither for bosons. Fermions are described by an [antisymmetric](#) wave function:

$$\psi(\dots, q_i, \dots, q_j, \dots) = -\psi(\dots, q_j, \dots, q_i, \dots).$$

Therefore, exchanging two fermions changes the sign of the wave function. In a path-integral or Monte Carlo description, different configurations can consequently contribute with different signs.

Let x denote a possible configuration, A_x the value of the observable for this configuration, and w_x its weight. The observable is formally written as

$$I = \sum_x w_x A_x.$$

For fermions, w_x is not necessarily positive. Thus, it cannot directly be interpreted as a probability. We instead sample with $|w_x|$ and keep the sign separately:

$$O = \left(\sum_x |w_x| \right) \langle \text{sign}(w_x) A_x \rangle_{|w|}$$

The notation $\langle \cdot \rangle_{|w|}$ means that the average is computed using probabilities proportional to $|w_x|$. The problem is that positive and negative contributions can almost cancel, leading to $\langle \text{sign}(w_x) \rangle_{|w|} \simeq 0$. The observable is then obtained as a small difference between large positive and negative contributions. Consequently, the numerical error becomes large, and a very large number of samples is required to obtain an accurate result. This is called the **fermions sign problem**.

5.2 QMD and identical particles

So far, the fiducial state has been chosen as

$$\phi_0 = \delta^{(dN)}(Q),$$

which is appropriate for distinguishable, Boltzmann particles. For identical particles, the fiducial state must be symmetrized or antisymmetrized. It is therefore replaced by

$$\phi_0(Q) = \frac{1}{N!} \sum_C \text{sgn}_\eta(C) \delta(CQ),$$

where C denotes a permutation of the N particles and CQ denotes the permuted configuration. The permutation factor is

$$\text{sgn}_\eta(C) = \eta^{\nu(C)},$$

where $\nu(C)$ is the parity of the permutation, that is, the number of pair exchanges needed to construct C . The parameter η is +1 for bosons, -1 for fermions.

The thermal density matrix is obtained after imaginary-time evolution:

$$\phi(Q, Q', \beta) = \langle Q | e^{-\beta \hat{H}} | Q' \rangle,$$

For an observable with a diagonal coordinate representation, the thermal average is

$$\langle O \rangle = \frac{\int_{\Omega^N} O(Q) \phi(Q, Q, \beta) dQ}{\int_{\Omega^N} \phi(Q, Q, \beta) dQ}.$$

The diagonal density matrix satisfies

$$\phi(Q, Q, \beta) = \langle Q | e^{-\beta \hat{H}} | Q \rangle \geq 0,$$

because the operator $e^{-\beta \hat{H}}$ is positive. Consequently, $\phi(Q, Q, \beta)$ can be interpreted as a probability density after normalization. Hence, if the classical [Molecular Dynamics algorithm can determine the exact diagonal thermal density matrix directly](#), the final Monte Carlo integration uses a non-negative weight and **does not suffer from sign cancellations!**

6 Conclusion

The paper presents a method for studying a quantum many-body system made of N particles by exploiting algorithms that are already used in Molecular Dynamics. The parameters q and p are evolved through Hamilton's equations, but the evolution is performed in imaginary time. In this way, the Molecular Dynamics evolution is used to reconstruct the thermal density matrix of the quantum system.

The density matrix starts from a delta-function initial condition and evolves according to the Bloch equation until the imaginary time reaches $\beta = \frac{1}{k_B T}$. At this point, the diagonal thermal density matrix is obtained. This quantity can then be used to compute thermal averages of observables with a Monte Carlo integration.

A central claim of the paper is that the method can avoid the fermion sign problem when measuring observables with a diagonal coordinate representation. In fact, the diagonal density matrix satisfies positivity, which means it can be normalized and therefore it can be used as a probability density in the final Monte Carlo integration. The efficiency of this reconstruction has to be tested numerically, especially when the number of particles increases or when the temperature becomes low.

The method also contains several possible sources of error. One comes from the choice of the fiducial state ϕ_0 , which is approximated by a sharply localized function.

Moreover, the continuous representation is generally inexact for non-quadratic potentials. Therefore, the discretization of the imaginary-time evolution, the numerical integration of Hamilton's equations, and the approximation of the correction term must all be controlled.

For observables with a diagonal coordinate representation, the proposed procedure gives a direct way to use Monte Carlo methods after the density matrix has been evolved. For non-diagonal observables, such as momentum distributions or dynamical correlation functions, the method does not yet provide a complete result. Additional work is necessary to reconstruct and sample the corresponding off-diagonal information.

In conclusion, the paper proposes an interesting connection between quantum imaginary-time evolution, Klauder's continuous representation, Molecular Dynamics, and Monte Carlo integration. Its main advantage is the possibility of using familiar Molecular Dynamics algorithms to evolve the parameters q and p of a quantum many-body system. Nevertheless, the method still requires a careful analysis of its approximation errors, computational cost, numerical stability, and actual efficiency for fermionic systems.

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